

Topic 10 — Electricity and Circuits

Edexcel GCSE (9-1) Physics (1PH0) — Paper 2

10.1 Basic Quantities

- **Current (I)** — rate of flow of charge; unit *ampere* (A).
- **Potential difference (V)** — work done per coulomb; unit *volt* (V).
- **Resistance (R)** — opposition to flow; unit *ohm* (Ω).
- **Charge (Q)** — measured in *coulombs* (C).

$$Q = I \times t \quad (\text{coulomb} = \text{ampere} \times \text{second})$$

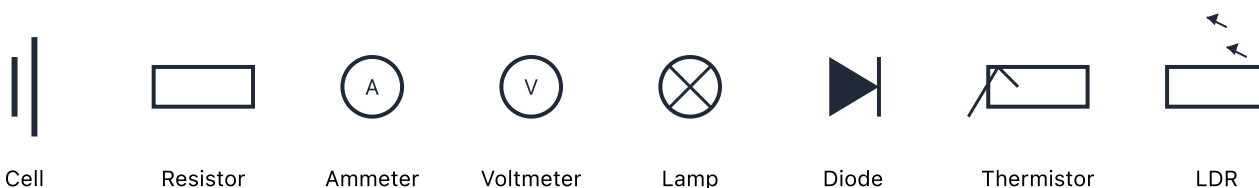
$$V = I \times R \quad (\text{Ohm's law})$$

$$P = V \times I = I^2 \times R = V^2 / R$$

$$E = V \times Q = V \times I \times t$$

10.2 Circuit Symbols

Common Circuit Symbols

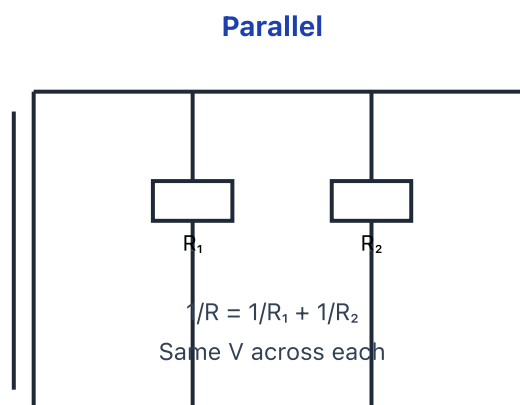
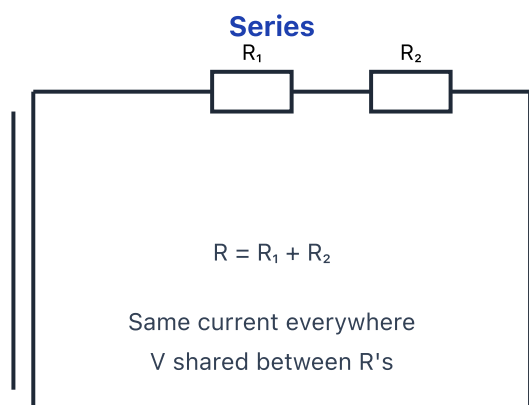


Ammeter in series

Voltmeter in parallel

Thermistor $R \downarrow$ as $T \uparrow$; LDR $R \downarrow$ as light $\uparrow$

10.3 Series and Parallel



10.4 I-V Characteristics

- **Fixed resistor** (constant T): straight line through origin — obeys Ohm's law.
- **Filament lamp**: curved S-shape — resistance increases with temperature.
- **Diode**: conducts one way only — nearly zero current below ~ 0.6 V forward; no current reverse.

10.5 Mains Electricity

UK mains: 230 V, 50 Hz **alternating current (AC)**.

Three-pin plug:

- **Live (brown)** — carries the alternating potential $\pm 230\text{ V}$.
- **Neutral (blue)** — 0 V ; completes circuit.
- **Earth (green/yellow)** — safety path; connects the metal case to ground.

A **fuse** melts if the current exceeds its rating, breaking the live line to protect the appliance.

Circuit breakers trip faster and can be reset.

10.6 The National Grid

Power stations generate at $\sim 25\text{ kV}$. **Step-up transformers** raise to 400 kV for efficient long-distance transmission: higher $V \rightarrow$ lower $I \rightarrow$ much smaller I^2R heat losses. Near towns, **step-down transformers** reduce voltage safely (e.g. $11\text{ kV} \rightarrow 230\text{ V}$).

Worked Example: A 2 kW kettle on 230 V draws $I = P/V = 2000/230 \approx \mathbf{8.7\text{ A}}$. Fuse rating: 13 A (just above 8.7 A).

Key Points Summary

- $V = IR$; $P = VI$; $Q = It$; $E = VIt$ — the essential equations.
- Series: add R 's. Parallel: reciprocal rule; lower combined R .
- Filament lamp and diode are non-Ohmic.
- Mains in UK = $230\text{ V AC } 50\text{ Hz}$; earth + fuse for safety.
- High-voltage transmission is efficient — transformers need AC.